MANIPAL UNIVERSITY JAIPUR
JAIPUR
(University under Section 2(f) of the UGC Act)

MANIPAL UNIVERSITY JAIPUR

Department Computer Science and Engineering

Course Hand-out

A. Basic Details:

Programme Name:	B.Tech-Computer Science and Engineering
Course Name:	Problem Solving Using Computers Lab
Course Code:	CSC1130
LTPC *(Lecture Tutorial Practical Credits):*	0 0 2 1
Session:	JULY 2025-NOVEMBER 2025
Class:	B. Tech 1st Year
Course Coordinator:	Dr. Rajat Goel
Course Instructor(s):	Dr. Jeyakrishnan V. | Dr Manoj Bohra | | Dr. Ashish Sharma | Dr. Siddhanta Kumar Singh | Dr Krati Dubey| Mr. Anurag Bhatnagar | Dr Suresh Kumar Jha| Dr. Bhawna Sharma | Dr. Virendra Kumar Meghwal |Mr. Arvind Kumar Mehta| Mr. Jay Shankar Sharma| | Dr. Shweta Gangrade | Dr. Vikas Kumar Boradak | Dr Arun Johar |Dr. Basudha Dewan| Ms. Tripti Kulshrestha| Dr. Atul Kumar Verma| Ms. Vipasha | Ms. Harshika Mathur | Dr. Princy Randhawa| Dr. Abhay Singh Bisht| Dr. Mohit Kushwaha
Additional Practitioner(s) – if any *(Industry Fellow/ Visiting Faculty/ Adjunct Faculty, etc.):*	To be identified and appointed later/ or at the beginning of the semester

B. Introduction:

Problem-Solving Using Computers Lab course focuses on basic computer fundamentals, programming fundamentals. By means of C language students learn to write a set of instructions to create a program so that the desired output can be generated by the computer.

C. Course Outcomes: At the end of the course, students will be able to

[CSC1130.1]. Design and develop flowcharts, algorithms, and pseudo-code to solve real-life problems.

[CSC1130.2]. Illustrate the basic programming concepts such as tokens, data types, operators, and control statements.

[CSC1130.3]. Demonstrate the concepts of array data type (1D and 2D), functions, structure, and union.

[CSC1130.4]. Describe the concept of pointers and file handling.

D. Program Outcomes and Program Specific Outcomes:

[PO.1]. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

[PO.2]. Problem analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using the first principles of mathematics, natural sciences, and engineering sciences.

[PO.3]. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

[PO.4]. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

[PO.5]. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

[PO.6]. The engineer and society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues, and the consequent responsibilities relevant to professional engineering practice.

[PO.7]. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

[PO.8]. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practices.

[PO.9]. Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

[PO.10]. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

[PO.11]. Project management and finance: Demonstrate knowledge and understanding of engineering. and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

[PO.12]. Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

E. Assessment Plan: Clearly write the criteria, its description, and associated marks for assessment of student achievements. In addition, the attendance requirements, assignments need to be mentioned in this section.

Criteria	Description	Maximum Marks
Lab Internal Assessment (Summative)	Continuous Assessments	**60** Continuous Assessment: 20 Lab record work: 15 Quiz: 20 (best 1 of 2) Attendance: 5
Lab End Term Examination (Summative)	End Term Lab Exam	**40** Viva:15 Writeup:10 Execution: 10 Lab Record Submission: 5
Total		100
Attendance (Formative)	A minimum of 75% Attendance is required to be maintained by a student to be qualified for taking the End Semester examination. The allowance of 25% includes all types of leaves including medical leaves.	
Homework/ Home Assignment/ Activity Assignment (Formative)	There are situations where a student may have to work in home, especially before a flipped classroom. Although these works are not graded with marks. However, a student is expected to participate and perform these assignments with full zeal since the activity/ flipped classroom participation by a student will be assessed and marks will be awarded.	

F. Syllabus: Digital computer fundamentals: Algorithms and flowcharts; Programming (Using C): data types, variables, operators, expressions, statements, control structures, functions, arrays and pointers, recursion, structures, files, input/output, some standard library functions.

References:

- E. Balagurusamy, “Programming in ANSI C”, 7th Edition, McGraw Hill Publication, 2017.
- Y. P. Kanetkar, “Let us C”, 16th Edition, BPB Publication, 2017.
- B. W. Kernighan, D. M. Ritchie, “The C Programming Language”, 2nd Edition, Prentice Hall of India, 2014.
- B. Gottfired, “Schaums Outline Series: Programming with C”, 3rd Edition, McGraw Hill Publication, 2012.

G. Lab Plan:

S.No.	Topics	Session Outcome	Mode of Delivery	Corres-ponding CO	Mode of Assessing CO
1	Algorithms and Flow Charts	Understand the flowcharts and design of algorithm	Demonstration	1130.1	Viva voce & End Term
2	Formula based C Programs	Understand the fundamentals of C programming.	Demonstration	1130.2	Viva voce & End Term
3	Control Structures: If statement	Choose the loops and decision-making statements to solve the problem.	Demonstration	1130.2	Viva voce & End Term
4	Control Structures: Switch	Choose the loops and decision-making statements to solve the problem.	Demonstration	1130.2	Viva voce & End Term
5	Control Structures: Loops	Choose the loops and decision-making statements to solve the problem	Demonstration	1130.2	Viva voce & End Term
6	1-D Array	Implement different Operations on arrays	Demonstration	1130.3	Viva voce & End Term
7	2-D Arrays	Implement different Operations on arrays	Demonstration	1130.3	Viva voce & End Term
8	Strings	Understand Strings	Demonstration	1130.3	Viva voce & End Term
9	Functions	Use functions to solve the given Problem	Demonstration	1130.3	Viva voce & End Term
10	Pointers	Understand pointers	Demonstration	1130.4	Viva voce & End Term
11	Structures	Understand structures and unions	Demonstration	1130.3	Viva voce & End Term
12	File Handling	Understand File-handling	Demonstration	1130.4	Viva voce & End Term

Outside Class Engagement Activities:

Type of Activity	**Difficulty Level**	**Hours Engaged**
Quiz 1	Medium	30 min-1 hour
Quiz 2	Medium	30 min-1 hour
End term Examination	Medium	2 hours
Hackathon/Industry Visit Other Events Participation	-	1-5 days

H. Course Articulation Matrix: Couse articulation matrix indicates the mapping of CO statement(s) to some PO statement(s). Here value 0 indicates no correlation, 1 as low correlation, 2 as moderate correlation, and 3 as substantial / strong correlation.

CO	STATEMENT	CORRELATION WITH PROGRAM OUTCOMES											
		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
1130.1	Design and develop flow charts, algorithms and pseudo code to solve real-life problems.	3											1
1130.2	Illustrate the basic programming concepts such as tokens, data types, operators, and control statements.	2											
1130.3	Demonstrate the concepts array data type(1D and 2D), functions, structure, and union.	2		1									2
1130.4	Describe the concept of pointers and file handling.	1											2